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NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Seventh Semester B.E. (Civil) (Credit System) Degree Examinations

Supplementary Examinations - December 2025

21CV702 / 20CV701 / 17CV701 - HYDROLOGY AND IRRIGATION ENGINEERING

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I** & **Unit – II** each and **One full** question from **Unit – III**.*

Unit – I		Marks	BT*	CO*	PO*
1. a)	Describe the practical applications of hydrology.	04	L*1	1	1
b)	Explain with a neat sketch the method of determining infiltration by double ring infiltrometer	06	L2	2	1
c)	There are four rain gauge stations existing in the catchment of a river. The average annual rainfall values at these stations are 800, 620, 400 and 540 mm respectively. i) Determine the optimum number of rain gauges in the catchment, if it is desired to limit the error in the mean value of rainfall in the catchment to 10% ii) How many more gauges will then be required to be installed?	10	L3	1	2
2. a)	Explain any 4 factors affecting Runoff.	04	L1	2	1
b)	Define Evapotranspiration. Explain Lysimeter method of estimating it	06	L2	2	1
c)	A 6h storm produced rainfall intensities of 7, 18, 25, 12, 10 and 3 mm/h in successive one hour intervals over a basin of 800 sq. km. The resulting runoff is observed to be 2640 hectare - metres. Determine ϕ -index for the basin.	10	L3	2	2
3. a)	Explain any 4 forms of precipitation.	04	L1	1	1
b)	Describe the construction and working of Symon's rain gauge.	06	L2	1	1
c)	A reservoir had an average surface area of 25km ² during June month. In that month the mean rate of inflow was recorded as 15m ³ /s, outflow 20 m ³ /s, monthly rainfall 12cm and change in storage 14×10 ⁶ m ³ . Assuming the seepage losses to be 1.5 cm, estimate the Evaporation in that month.	10	L3	2	2
Unit – II					
4. a)	Define irrigation and explain the necessity of irrigation in India	04	L1	3	1
b)	The gross commanded area for a distributary is 6000 hectares, 80% of which is culturable irrigable. The intensity of irrigation for Rabi season is 50% and that for Kharif season is 25%. If the average duty at the head of the distributary is 2000 hectares/cumec for Rabi season and 800 hectares/cumec for Kharif season, find out the discharge required at the head of the distributary from average demand considerations.	06	L3	2	2
c)	Explain the three ways of aligning an irrigation canal	10	L2	4	2
5. a)	Define Duty, Delta and Base period. Give the relation between the same.	04	L1	3	1
b)	Explain the different irrigation efficiencies.	06	L2	3	1

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21CV702 / 20CV701 / 17CV701**Supplementary - Dec. 2025**

c)	Design a regime channel for a discharge of 50 cumecs & silt factor 1.1, using Lacey's Theory.	10	L3	4	3
6. a)	Explain sprinkler irrigation system.	04	L1	3	1
b)	Explain in brief flooding method of surface irrigation system.	06	L2	3	1
c)	Design an irrigation channel to carry 40 cumec of discharge, with B/D that i.e., base width to depth ratio as 2.5. The critical velocity ratio is 1.0. Assume a suitable value of Kutler's rugosity coefficient and use Kennedy's method.	10	L3	4	3

Unit – III

7. a)	Explain with neat sketch the storage zones of a Reservoir	10	L2	5	1
b)	Explain the mass curve method that can be used for determining: Reservoir capacity for fulfilling given demand.	10	L2	5	2
8. a)	Explain briefly with neat sketches any 4 different forces acting on gravity dam.	10	L2	5	1
b)	Explain in brief the Hydraulic failure of an Earthen dam.	10	L3	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
